CAMPUS360

ServiceNow scoped application · v2.0.0

User Manual & Interview Demo Guide

University IT Issue Tracking & Campus Management System
Native ServiceNow UI (PDI) and the Campus360 Portal

Inside

- Project status: what is done and what is next
- One-time demo setup (users and sample data)
- Step-by-step demo of every module in the PDI
- Step-by-step demo of the portal for every role
- A 15-minute interview demo script and talking points
- Security, testing, Q&A and troubleshooting

Instance dev382380 · scope x_774308_campus360 · November 2026

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1. Is the project complete?

Short answer: the delivered scope is complete, deployed and tested. Some planned automation (Parts 4-9) was intentionally not started.

1.1 What is finished and live on your instance

Area	Status	Evidence
Data model: 26 Campus360 tables (IT issues, hostel, academics, attendance, labs, registration, onboarding)	Done, deployed	Visible under Campus360 in the navigator
Security: 13 roles, 9 groups, 421 ACLs, row-level rules, personal-data protection (D-2), query rules (W-3)	Done, deployed	ATF security suites pass
Workflow guards for IT issues and hostel complaints (native form, list edit, REST, scripts)	Done, deployed	2 guard business rules + field ACLs
Native (PDI) application menu: 8 sections, 40 modules	Done, deployed	Filter navigator: Campus360
Campus360 Portal (React UI page) with role-based screens and "Campus records"	Done, deployed	https://dev382380.service-now.com/x_774308_campus360_portal.do
Portal REST API: 20 routes, GlideRecordSecure only	Done, deployed	Scripted REST API "Campus360 Portal API"
Automated tests: 4 ATF tests (111 steps) + 270 local unit/build tests	All passing	Section 9 of this manual
Documentation (architecture, schema, roles, guides, reports)	Done	docs/ folder and GitHub
Source code on GitHub	Pushed	github.com/meesalavamsi/University_IT_Issue_Tracking_System_With_Service_Now

1.2 What is NOT done (and why that is fine for an interview)

Planned item	Part
Priority from impact x urgency, auto assignment rules, auto-close of resolved issues, notifications, SLAs, escalate to Incident	Part 4
Registration approval flow by HOD, student provisioning	Part 5
Faculty onboarding flow, task generation from templates, laptop allocation	Part 6
Roster generation, attendance lock, attendance % roll-up, missing-attendance jobs	Part 7
Room allocation request flow, occupancy reconciliation	Part 8
Preventive-maintenance generator, maintenance from IT issue	Part 9
Service Catalog items and a Service Portal (/sp) version of the screens	Later

BE HONEST ABOUT THIS

The tables, roles and ACLs for these features already exist; only the automation (Flow Designer flows, scheduled jobs, notifications) is pending. In an interview, present them as your **roadmap**: "the foundation and security are production-grade; next phases add automation."

W-4 (audit history visibility) was investigated and deliberately left open: it concerns global platform tables (sys_audit / sys_history_line) that a scoped app should not modify. Auditing stays ON.

1.3 The one thing you MUST do before a demo

Your instance currently has **no sample business data** (0 students, 0 hostels, 0 issues). Only the 8 IT issue categories and 24 subcategories are loaded. This is by design: the app was installed without demo data. If you open the portal now, most screens will say "nothing here yet". Follow **Section 3 (one-time demo setup, about 30-40 minutes)** once, and every screen in this manual will have something to show.

> **TIP**
> Your PDI (Personal Developer Instance) goes to sleep after inactivity and is reclaimed if unused for a long time. Log in to developer.servicenow.com a day before the interview, wake the instance, and run through the demo once.

2. Campus360 in one page

Use this page to open your interview. Memorise the 30-second pitch.

SAY THIS IN THE INTERVIEW

"Campus360 is a scoped ServiceNow application that started as a University IT issue tracker and grew into a campus management system. Students, faculty, wardens, technicians, the registrar and leadership all use one platform. It has 26 custom tables, 13 roles and 421 ACLs, with row-level and field-level security so that, for example, a warden only sees residents of their own hostel and personal data is visible only to the registrar. There are two interfaces: the native ServiceNow back-office, and a modern React portal I built on a UI page that talks to a secured Scripted REST API. Everything is source-controlled with the ServiceNow SDK (Fluent), covered by 111 ATF steps and 270 unit tests, and deployed with a repeatable pipeline."

2.1 Key facts

Item	Value
Instance / scope	dev382380 (PDI) · scope x_774308_campus360 · version 2.0.0
Built with	ServiceNow SDK 4.14 (Fluent, TypeScript), React 18 for the portal
Tables	26 (3 extend Task: IT Issue, Hostel Complaint, Maintenance Request)
Roles / groups	13 scoped roles · 9 assignment groups
Security	421 ACLs, 6 before-query rules, 2 workflow-guard rules, C360Security script include
Business rules	9 (2 guards, 6 query restrictions, 1 registrar-review stamp)
Navigator	Application "Campus360": 8 sections, 40 modules + "Campus360 Portal" link
Portal	UI page x_774308_campus360_portal.do + Scripted REST API (20 routes)
Tests	4 ATF tests / 111 steps · 270 local tests (node --test)

2.2 Architecture (how the pieces fit)

Layer	What it is	Where to show it
Users & roles	sys_user, 9 C360 groups grant the roles (student/faculty roles given directly)	User Administration > Groups
Native UI	Lists and forms for every table, grouped in the Campus360 menu	Filter navigator: Campus360
Portal UI	React single-page app served by a UI page; role-based menu; same data	Campus360 > Campus360 Portal
API	Scripted REST API (/api/x_774308_campus360/portal/...), every route needs login + Campus360 role	System Web Services > Scripted REST APIs
Server logic	TypeScript modules compiled to scoped server code; only GlideRecordSecure (ACLs always apply)	Studio / source on GitHub
Security	Table, field and row ACLs + before-query BRs + workflow-guard BRs	System Security > Access Control (ACL)
Data	26 scoped tables, audited where personal data lives	System Definition > Tables

BE HONEST ABOUT THIS
You may be asked "Is this a Service Portal?" The answer: **No, it is a custom React portal on a UI page** (the modern "UI page + REST API" pattern, similar to how Now Experience / UI Builder apps separate front end and back end). A classic Service Portal (/sp with widgets) was planned but not built. The advantage of this design is that the browser never decides permissions: the REST API runs every query through GlideRecordSecure, so the same ACLs protect both UIs.

2.3 The 13 roles

Role	Contains	Who / what they do
user	-	Base access: portal, own records
student	user	Raises requests, sees own profile, hostel, attendance
faculty	user	Own sections, class sessions, marks attendance
hod	faculty	Department oversight, registrations (read-only)
technician	user	Works IT issues of their own support groups
it_manager	technician	Whole IT queue, categories, technicians
warden	user	Own hostel: rooms, residents, complaints
maintenance	user	Facilities work orders
lab_admin	user	Laboratories, PM plans, lab assets
registrar	user	Students, academics, registrations, student personal data
faculty_admin	user	Faculty records, onboarding, faculty personal data
leadership	user	Read-only analytics, no personal fields
admin	all staff roles	Campus360 application administrator

All roles are named `x_774308_campus360.<role>`. The 9 groups: C360 IT Service Desk, Desktop & Hardware Support, Network & Wi-Fi Team, Lab Support (technician); Facilities & Electrical (maintenance); Hostel Administration (warden); Registrar Office (registrar); Faculty Onboarding Team (faculty_admin); IT Management (it_manager).

3. One-time demo setup

Do this once as admin in the native UI. Afterwards every demo in this manual works. Use fictional names only.

3.1 Log in

1. Open **https://dev382380.service-now.com** and log in as **admin** (your PDI admin password from developer.servicenow.com).
2. Your admin user already has every Campus360 staff role, including *admin*, so it can see and create everything.
3. In the left **All** menu, type **Campus360** in the filter. You should see the sections IT Issues, Students, Faculty, Academics, Attendance, Hostel, Labs & Infrastructure, Administration.

3.2 Create the demo users

Go to **All > User Administration > Users > New**. Create each user below (User ID, First name, Last name; tick **Active**). You do not need passwords: you will use **Impersonate user**.

User ID	Name (example)	Give access by	Demo persona
demo.student	Asha Student	Role: x_774308_campus360.student (Roles tab > Edit)	Student
demo.student2	Rahul Student	Role: x_774308_campus360.student	Second student (security demo)
demo.faculty	Priya Faculty	Role: x_774308_campus360.faculty	Teacher / attendance
demo.warden	Suresh Warden	Group: C360 Hostel Administration	Hostel warden
demo.tech	Kiran Tech	Group: C360 Network & Wi-Fi Team	Technician (Wi-Fi)
demo.tech2	Anil Tech	Group: C360 Desktop & Hardware Support	Technician, other group
demo.registrar	Leela Registrar	Group: C360 Registrar Office	Registrar office
demo.leader	Dean Leader	Role: x_774308_campus360.leadership	Leadership / analytics
demo.facilities	Manoj Facilities	Group: C360 Facilities & Electrical	Maintenance staff

To add to a group: open the user > **Groups** related list > **Edit** > move the group to the right > Save. To add a role: **Roles** related list > **Edit**.

> **TIP**
> Granting staff roles through groups (not directly) is the design rule of this app. Mention it: "roles follow the group, so onboarding is one step."

3.3 Create the reference data (in this order)

All of these are in the **Campus360** menu. Open the list, click **New**, fill the fields marked with a red asterisk, and **Submit**.

#	Menu path	What to create (example values)
1	Academics > Departments	Code CSE, Name Computer Science and Engineering
2	Academics > Academic Terms	Name "Odd Semester 2026", Academic Year 2026-27, start/end dates around today, tick Current Term and Active

#	Menu path	What to create (example values)
3	Students > Programs	Code BTECH-CSE, Name B.Tech CSE, Department CSE, Duration 4, Total Semesters 8
4	Academics > Courses	Code CS301, Name Database Systems, Credits 4, Department CSE
5	Faculty > Faculty	User demo.faculty, Employee ID F1001, Department CSE, Status Active
6	Students > Students	User demo.student, Student ID 22CSE001, Department CSE, Program B.Tech CSE, Advisor Priya Faculty, Status Active. Repeat for demo.student2 (22CSE002)
7	Academics > Course Sections	Course CS301, Term Odd Semester 2026, Section A, Instructor Priya Faculty, Max Students 60, Status Active
8	Academics > Enrollments	Student Asha -> CS301-A (Enrolled). Repeat for Rahul
9	Attendance > Class Sessions	Course Section CS301-A, Date = today (and one for yesterday), Start 10:00, End 11:00
10	Labs & Infrastructure > Laboratories	Code LAB-DB, Name Database Lab, pick any Location
11	Hostel > Hostels	Code H1, Name Krishna Hostel, Type Boys/Co-ed, Building = any location, Warden Suresh Warden
12	Hostel > Rooms	Hostel H1, Room Number 101, Room Type Double, Bed Capacity 2 (add 102 too)
13	Hostel > Allocations	Student Asha, Room H1-101, Check-in Date today, Status Active
14	Students > Registrations - All	New registration: First/Last name, Email (fictional, e.g. applicant@example.edu), leave state Submitted

TIP
Order matters because of references (a Student needs a Department and Program; a Section needs a Course, Term and Instructor). Without step 13 (allocation) the student cannot raise a hostel complaint: that is a security rule, not a bug.

3.4 Create a few sample tickets

1. Impersonate **demo.student** (Section 3.5), open the portal, raise an IT issue: category *Network / Wi-Fi* type, title "Wi-Fi drops in hostel block A". Raise a hostel complaint and a second IT issue (e.g. hardware: "Projector not working in room 204").
2. End impersonation. As admin, open each new IT issue (IT Issues > Open) and set **Assignment group**: Wi-Fi issue -> C360 Network & Wi-Fi Team; projector -> C360 Desktop & Hardware Support. (Auto-assignment is a Part 4 feature, so you do it by hand.)
3. Impersonate **demo.facilities** or use admin to create one maintenance request (Labs & Infrastructure > Maintenance Requests - Open > New) for the Database Lab.

3.5 How to impersonate (switch persona without passwords)

1. Click your **profile avatar** (top right) > **Impersonate user**.
2. Type the user (e.g. demo.warden) and select it. The page reloads as that user.
3. Open the portal with the menu **Campus360 > Campus360 Portal** or the URL `https://dev382380.service-now.com/x_774308_campus360_portal.do`.
4. To go back: avatar > **End impersonation** (or impersonate System Administrator).

TIP

Before the interview, open two browser windows: one normal (admin) and one private window for the portal. Rehearse each persona switch so it takes under 10 seconds.

4. Demo A: Campus360 in the PDI (native ServiceNow UI)

This shows the back-office side: lists, forms, the data model and the platform configuration. Log in as admin.

4.1 The application menu

1. Click **All** and type **Campus360**.
2. Show the 8 sections and explain that each module has roles: users only see the sections for their job.
3. Impersonate demo.warden and filter Campus360 again: only the **Hostel** section and the Portal link appear. End impersonation.

> **SAY THIS IN THE INTERVIEW**
> "The navigator itself is role-aware. 40 modules, each restricted by role, so a warden sees only Hostel, and a student sees only the portal."

4.2 IT Issues

Module	What to show
Create New	The IT issue form (extends Task): Category drives Subcategory; lab categories make Laboratory required. Number prefix ITI.
Open / Open - Unassigned	Active issues; the unassigned filter is the dispatcher view
My Groups' Work / Assigned to Me	Dynamic filters: same module, different result per technician
Resolved / All	Resolved = state 6; All includes closed and cancelled
Technicians	Technician profiles (IT manager)
Issue Categories / Subcategories	8 categories, 24 subcategories pre-loaded; flags requires_lab / requires_room change the form

States: 1 New, 2 In Progress, -5 On Hold, 6 Resolved, 3 Closed, 7 Cancelled. Resolve needs a resolution code and notes; hold needs a reason.

Security moment (very strong in interviews)

1. Impersonate **demo.tech2** (Desktop & Hardware). Open the **Wi-Fi** issue (assigned to Network & Wi-Fi).
2. Show that State, Assignment, Work notes are **read-only**: a technician can only work issues of their own group.
3. Open the **projector** issue (their group): the fields are editable. End impersonation.

> **SAY THIS IN THE INTERVIEW**
> "Field ACLs with conditions enforce group authority, and a before-update guard business rule blocks the same change from scripts, list edit or REST, so there is no back door."

4.3 Students and Academics

Module	What to show
Students > Students	Student profile: ID, department, program, semester, advisor, and personal fields (DOB, phone, address, guardian) visible to admin/registrar
Registrations - Pending / All	Admission applications; states Submitted -> Registrar Review -> Pending HOD -> Approved/Rejected -> Enrolled
Programs, Departments, Academic Terms, Courses	The academic structure
Course Sections	Course + term + section + instructor; unique index stops duplicate sections
Enrollments	Student in a section; holds sessions held/attended, attendance %, exam eligibility

Personal-data moment

1. As admin, open Asha's student record and fill **Date of Birth** and **Personal Phone**. Save.
2. Impersonate **demo.faculty** and open `x_774308_campus360_student_list.do` (paste after the instance URL). Faculty sees only students enrolled in their sections, and the personal fields are **hidden**.
3. Impersonate **demo.student2**, open the same URL: Rahul sees only his own record. End impersonation.

> **SAY THIS IN THE INTERVIEW**
> "Row-level ACLs answer 'which records', field ACLs answer 'which columns'. I also added query_range and query_match deny ACLs so nobody can guess a birth date by filtering on it."

4.4 Faculty and Attendance

- **Faculty**: staff directory; personal fields only for faculty_admin. **Onboarding / Onboarding Tasks / Templates**: tables and security ready; the flow is Part 6.
- **Class Sessions - Today / Class Sessions**: timetable instances. **Student Attendance**: one row per student per session (unique index, cannot be deleted even by admin).
- **Low Attendance (< 75%)**: enrollments under the threshold. **Faculty Attendance**: staff self check-in table.

4.5 Hostel

- **Hostels** (warden, assistant warden, capacity), **Rooms**, **Allocation Requests** (status requested), **Allocations**, **Complaints - Open** (number prefix HCP).
- Impersonate demo.warden: they see only **their** hostel's rooms, allocations and complaints. Guardian name/phone are visible to the warden only for active residents (emergency need).

4.6 Labs & Infrastructure and Administration

- **Laboratories**, **Maintenance Requests - Open / All** (MNT prefix, states New, Scheduled, In Progress, On Hold, Completed, Cancelled), **Preventive Maintenance Plans**, **Campus Hardware Assets** (platform alm_hardware, shown not modified).
- **Administration > Properties**: 10 x_774308_campus360.* properties (e.g. security.debug). **Campus360 Groups**: the 9 C360 groups.

4.7 Behind the scenes (for technical interviewers)

Show	Where
The app record and all its files	All > System Applications > My Company Applications > Campus360 (or Studio)
Tables and columns	System Definition > Tables, filter name starts with x_774308_campus360
ACLs (421)	System Security > Access Control (ACL), filter name starts with x_774308_campus360 (elevate to security_admin if needed)
Business rules (9)	System Definition > Business Rules, filter name starts with C360
Script include C360Security	System Definition > Script Includes
Scripted REST API	System Web Services > Scripted REST APIs > Campus360 Portal API (20 resources)
UI page (portal host)	System UI > UI Pages > x_774308_campus360_portal
ATF tests	Automated Test Framework > Tests, names start with C360

SAY THIS IN THE INTERVIEW

"I did not click these together in Studio. They are all TypeScript source files built and deployed with the ServiceNow SDK, so the whole app is in Git, reviewed and repeatable."

5. Demo B: the Campus360 Portal

The end-user experience. Open it from Campus360 > Campus360 Portal, or https://dev382380.service-now.com/x_774308_campus360_portal.do

5.1 Common to everyone

- A short **splash screen**, then a gradient **sidebar** with only the screens your role allows, a top bar with breadcrumb, date and your user pill with role chips.
- **Dashboard**: hero greeting, quick actions, recent activity and role-specific panels with animated counters.
- **Raise a request** (3 steps) and **My requests** for every user.
- **Campus records** (staff only): the same 40 modules as the native navigator, grouped and filtered by role, opened inside the portal.
- On a phone-width window the sidebar collapses behind a **Menu** button.

5.2 Student (impersonate demo.student)

1. **Dashboard**: My profile (student ID, program, department, semester, advisor), My hostel (room/bed), My attendance per section (red below 75%), recent requests.
2. **Raise a request** > IT issue > pick category and subcategory, location, urgency, title (min 5 chars) > **Review** > **Submit**. Show the number (ITI...) and open it.
3. **Raise a request** > Hostel complaint: the hostel is taken automatically from the student's allocation (cannot be spoofed).
4. **My requests**: all IT issues, complaints and maintenance requests raised by me with state and last update.
5. **Request details**: progress timeline (New > In Progress > Resolved > Closed), add a comment and an attachment.

> **SAY THIS IN THE INTERVIEW**
> "The student never chooses the caller or the hostel. The server sets them from the logged-in user, so nobody can raise a ticket on behalf of someone else."

5.3 Technician (impersonate demo.tech)

1. Open **IT work queue**: counters for New / In progress / On hold / Older than 3 days, status filter and search.
2. Open the Wi-Fi issue > **Assign to me** > **Start work** (state In Progress).
3. **Put on hold** > reason "Awaiting parts" > **Resume**.
4. **Resolve** > resolution code (e.g. Configuration change) + notes. Then show **Reopen**.
5. Show **Cancel issue** needs a work note (5+ chars) and confirmation. **Add work note** adds a journal entry.
6. Only buttons valid for the current state appear; the server re-checks every action.

Then impersonate **demo.tech2**: the Wi-Fi issue is not in their queue. IT managers (e.g. admin) see every open issue.

5.4 Warden (impersonate demo.warden)

1. **My hostel**: hostel occupancy, rooms, current residents (guardian contact for own residents only).
2. Open complaints with state buttons: **Start** (New), **Hold**, **Resume**, **Resolve** (asks for resolution notes). **Open** shows the full record.
3. **Maintenance** and **Campus insights** are also available; **Campus records** shows only the Hostel section.

5.5 Faculty (impersonate demo.faculty)

1. **Classes & attendance**: sections I teach and sessions from the last 14 days.
2. **Take attendance** on today's session: **All present**, then change one student to Absent or Late. The page counts unsaved changes.
3. **Save**. Save again: the marks are updated, never duplicated (unique index + upsert).
4. Optionally show the result in the native list Attendance > Student Attendance as admin.

BE HONEST ABOUT THIS
The attendance **percentage** on the enrollment and student dashboard is a stored field whose automatic roll-up is Part 7, so it will not change after you mark attendance. Say so if asked.

5.6 Registrar (impersonate demo.registrar)

1. **Registrations**: submitted and in-review applications.
2. Open the one you created: tick **Documents verified**, choose **Recommendation = recommend**, **Save review**. State moves to Pending HOD approval.
3. Show the rules: recommend is refused unless documents are verified; reject needs a reason. A business rule stamps "reviewed by" and "reviewed at" automatically.

5.7 Leadership (impersonate demo.leader)

Campus insights: counts and breakdowns of IT issues (by state, priority, category), complaints, maintenance, registrations and hostel occupancy, computed **only from records the viewer may see**, with no personal data.

SAY THIS IN THE INTERVIEW
"Even the statistics respect security. Counts come from before-query business rules that mirror the read ACLs, so aggregate numbers cannot leak rows a user is not allowed to see."

5.8 Campus360 admin (as admin)

- **Administration**: row counts per Campus360 table, links to native lists and a list explorer.
- **Campus records**: all 8 sections / 40 modules. Lists open inside the portal with search, count, refresh and **New**; a record opens as a form with a link to the classic form; platform tables (hardware assets, properties, groups) open the native UI in a new tab.

5.9 Maintenance / facilities (impersonate demo.facilities)

Maintenance: open maintenance requests they may see; **New work order** opens the request form (laboratory or hostel, maintenance type, urgency). Status changes happen in the full record.

6. Suggested 10-15 minute interview demo flow

Min	Step	Persona
0-1	30-second pitch (Section 2) + architecture table	-
1-3	Native menu: show the 8 sections; impersonate warden to show role-aware navigator	admin -> warden
3-5	Student raises a Wi-Fi IT issue and a hostel complaint in the portal; My requests	demo.student
5-7	Technician works the issue in the IT work queue: assign, start, hold, resolve	demo.tech
7-8	Wrong-group technician cannot change it (native form read-only fields)	demo.tech2
8-9	Warden resolves the hostel complaint	demo.warden
9-10	Faculty takes attendance	demo.faculty
10-11	Student sees request resolved with timeline	demo.student
11-12	Personal data: faculty sees student list without DOB/phone	demo.faculty
12-13	Leadership insights	demo.leader
13-15	Behind the scenes: SDK source on GitHub, ACL list, ATF results	admin

TIP
If time is short, do only: pitch -> student raises issue -> technician resolves -> wrong-group technician blocked -> student sees it resolved. That story shows the full lifecycle and the security.

7. Process reference (know these by heart)

7.1 IT issue lifecycle

Action	From state	To state	Required
Start	New	In Progress	Assigns to me if unassigned
Hold	New, In Progress	On Hold	Reason: awaiting caller / parts / vendor / scheduled visit
Resume	On Hold	In Progress	-
Resolve	New, In Progress, On Hold	Resolved	Resolution code + close notes (5+ chars)
Reopen	Resolved	In Progress	-
Cancel	New, In Progress, On Hold	Cancelled	Work note explaining why (5+ chars) + confirmation
Close	Resolved	Closed	Manual in native form (auto-close is Part 4)

Every portal action is checked on the server: record visible (else 404), record writable, operational authority (IT manager/admin or technician in the assignment group; having raised the ticket never counts), and every changed field writable. Any failure returns 403 and nothing changes.

7.2 Hostel complaint

Same states. Warden actions: Start, Hold, Resume, Resolve (notes required). Only the warden or assistant warden **of that hostel** (or Campus360 admin) can act. Students can complain only about the hostel of their own active allocation. Confidential complaints (is_confidential) are visible only to the caller, the hostel warden and admin.

7.3 Maintenance request

States: New, Scheduled, In Progress, On Hold, Completed, Cancelled. Created from the portal (corrective) or native form; worked in the native form by the assigned facilities group.

7.4 Student registration

Submitted -> (more info) Registrar Review -> (recommend, documents verified) Pending HOD Approval -> Approved / Rejected -> Enrolled. Registrar review is in the portal; HOD approval and provisioning are Part 5.

7.5 Attendance

Instructor or co-instructor opens a session from the last 14 days, marks present / absent / late / excused / medical / on duty, saves. One record per student per session (unique index). Attendance records cannot be deleted; corrections are status changes and stay in the audit history.

8. Security design (the part interviewers love)

Control	How it works	Demo
Role-based menus	Every module has roles	Impersonate warden: only Hostel
Row-level ACLs	Script ACLs via C360Security (e.g. faculty sees only enrolled students, warden only own residents)	Faculty opens student list
Field-level ACLs	Personal fields readable only by self, registrar/faculty_admin, admin	DOB hidden for faculty
Blind-query protection	query_range / query_match deny ACLs on sensitive fields	Explain
D-2 fix	Platform admin alone does NOT see personal data; needs explicit Campus360 role	Explain
Before-query BRs (W-3)	Lists, reports and counts only include rows the user may read	Leadership insights
Workflow guards	Conditional field ACLs + before-update guard BRs, for every interface	tech2 blocked
Secure API	All 20 routes need authentication + Campus360 role ACL; GlideRecordSecure only; the browser never authorises	Explain
Auditing	audit=true on personal and attendance tables; attendance cannot be deleted	Record history
No web-service access	allowWebServiceAccess=false on personal tables	Explain

9. Showing the tests

9.1 ATF (Automated Test Framework) on the instance

1. As admin, open **System Properties** (sys_properties.list), find `sn_atf.runner.enabled` and set it to **true**.
2. Go to **Automated Test Framework > Tests** and filter names starting with **C360**. You will see 4 tests:
3. Open one and click **Run Test**. ATF creates its own temporary users and data and rolls them back afterwards.
4. After the demo set `sn_atf.runner.enabled` back to **false**.

- C360 Security - Student profile access regression (Part 3)
- C360 Integrity - Attendance and escalation regression
- C360 Security - Native workflow boundaries and statistics (rev 3)
- C360 Portal - API end-to-end regression

Last full run: **111 of 111 steps passed**.

9.2 Local tests (on your laptop)

```
cd University_IT_Issue_Tracking_System_ServiceNow
npm install
npm test          # 270 tests, all passing
npm run build     # compiles the app with the ServiceNow SDK
```

> **BE HONEST ABOUT THIS**
> Do not run `npm run deploy` just to demo; it reinstalls into the instance. Never uninstall the app: that deletes the data.

10. Likely interview questions and good answers

Q. Why a scoped application?

Isolation and safety: everything lives in x_774308_campus360, can be versioned, installed and upgraded as one unit, and cannot accidentally change global tables. Campus360 never modifies platform ACLs or tables.

Q. Why extend the Task table for IT issues, complaints and maintenance?

Task gives states, assignment group/assigned to, work notes and comments (journals), approvals, SLA readiness and standard list/form behaviour for free.

Q. How do you stop a technician from closing a ticket in another group?

Three layers: conditional field ACLs on the workflow fields (UI, list edit, REST), a before-update guard business rule for scripted updates, and the portal's own server-side check. The guard judges authority on the record before the change.

Q. What is the difference between ACLs and before-query business rules?

ACLs decide whether a row or field can be read or written; a before-query BR adds a filter to the query so lists, reports and counts never include rows you cannot read (no "rows removed by security" gaps and no leaking counts).

Q. Why GlideRecordSecure in the REST API?

GlideRecord runs as system and ignores ACLs; GlideRecordSecure enforces the caller's ACLs, so the API can never return more than the native UI would.

Q. What was the D-2 issue?

Personal-data field ACLs were role-based, and platform admin implicitly passes role checks. I changed 73 field ACLs to require explicit Campus360 role membership, so a platform admin without a Campus360 role sees no personal fields.

Q. How did you build and deploy it?

ServiceNow SDK (Fluent) in TypeScript, build with now-sdk build, deploy with now-sdk install (no demo data), take a baseline download before each deploy, then run ATF and unit tests. Source on GitHub.

Q. What bugs did you find in testing?

Examples: on this instance an empty GlideElement is falsy, so a null check had to replace a truthiness check; journal fields cannot be set with setValue and setJournalEntry is blocked in a scope, so work notes are written with property assignment. ATF caught both.

Q. Is the portal a Service Portal?

No, a React app on a UI page with a Scripted REST API. It gives full control of the UX, and all authorisation stays on the server. A Service Portal or Employee Center version is a possible next step.

Q. What would you add next?

Part 4: priority matrix, assignment rules, notifications, SLAs and auto-close in Flow Designer; then registration approval, onboarding flows, attendance roll-up and a preventive-maintenance generator.

Q. How do you handle personal data / privacy?

Least privilege: field ACLs, deny query ACLs, no web-service access on personal tables, masked list layouts, auditing on, and leadership analytics use non-sensitive fields only.

Q. How many users can this support?

The data model is indexed on the hot paths (unique indexes on enrollment, attendance, rooms; indexes on warden, faculty, term), and every query is filtered server-side, so it scales with the platform.

11. Troubleshooting

Problem	Fix
Instance URL does not load / "hibernating"	Log in to developer.servicenow.com and wake the instance (takes a few minutes).
Portal says "Campus360 access not set up"	The user has no Campus360 role. Add the role or group (Section 3.2) and reload.
Portal screens are empty	No sample data yet: do Section 3.3-3.4.
Student profile blank on dashboard	Student record missing or its User field not set to that sys_user.
"Hostel complaint refused"	The student needs an Active room allocation (Hostel > Allocations).
Technician does not see an issue	Set the issue's Assignment group to a group the technician is in.
Faculty sees no sessions	Section must have Instructor = their faculty record, and sessions dated within the last 14 days.
Warden sees nothing	Hostel.Warden must be that user AND the user must be in C360 Hostel Administration.
401/403 in portal	Session expired or missing role: reload and log in again.
A field is blank	A field ACL is hiding it on purpose (personal data).
New role not applied	Log out/in, or end and restart impersonation.

Appendix A. Native navigator modules (40)

Section	Modules
IT Issues	Create New, Open, Open - Unassigned, My Groups' Work, Assigned to Me, Resolved, All, Technicians, Issue Categories, Issue Subcategories
Students	Students, Registrations - Pending, Registrations - All, Programs
Faculty	Faculty, Onboarding, Onboarding Tasks, Onboarding Templates
Academics	Departments, Academic Terms, Courses, Course Sections, Enrollments
Attendance	Class Sessions - Today, Class Sessions, Student Attendance, Low Attendance (< 75%), Faculty Attendance
Hostel	Hostels, Rooms, Allocation Requests, Allocations, Complaints - Open
Labs & Infrastructure	Laboratories, Maintenance Requests - Open, Maintenance Requests - All, Preventive Maintenance Plans, Campus Hardware Assets
Administration	Properties, Campus360 Groups

Portal screens by role

Screen	Who sees it
Dashboard, Raise a request, My requests	Everyone with a Campus360 role
IT work queue	technician, it_manager, admin
My hostel	warden
Maintenance	maintenance, lab_admin, warden, technician, admin
Classes & attendance	faculty, hod
Registrations	registrar, hod, admin
Campus insights	leadership, admin, it_manager, hod, registrar, warden, technician, lab_admin, maintenance
Administration	admin
Campus records	Staff roles, filtered to the modules they have in the navigator

Useful links

What	Link
Instance	https://dev382380.service-now.com
Campus360 Portal	https://dev382380.service-now.com/x_774308_campus360_portal.do
Source code	https://github.com/meesalavamsi/University_IT_Issue_Tracking_System_With_Service_Now
Detailed docs (in repo)	docs/ : ARCHITECTURE, DATABASE_SCHEMA, ROLE_PERMISSION_MATRIX, CAMPUS360_END_USER_GUIDE, CAMPUS360_ADMINISTRATOR_GUIDE, WORKFLOW guide, DEPLOYMENT_EXECUTION_REPORT